Period Obstructions for Weighted Binary Expansions

Denominator valuations and a 1990 periodicity question

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Abstract

For binary digits ε_n, consider the weighted value $X = \sum_{n\geq 1} n\varepsilon_n/2^n$. We prove that an eventually periodic representation with a nonconstant tail forces an odd prime square in the reduced denominator of X. Consequently, no rational with a nontrivial squarefree odd denominator part has any eventually periodic representation; in particular this holds for $1/3$. This rules out the universal periodicity possibility raised by Borwein and Loring in 1990. More precisely, an exact valuation identity and a finite closure under prime divisors of multiplicative orders give an explicit integer D_Q, depending only on the odd denominator Q of X, such that every ordinary periodic-tail denominator divides D_Q and every minimal period divides $\mathrm{ord}_{D_Q}(2)$. We give full proofs, a general-base extension, boundary examples, and an exact reproducible verifier. This does not resolve Erdős Problem #261; historical priority of the additional results is not certified by the bounded literature search reported here.

1 The historical question and the auxiliary problem

Borwein and Loring call an expression

$$x = \sum_{n\geq 1} \frac{n\varepsilon_n}{2^n}, \qquad \varepsilon_n \in \{0, 1\}, \tag{1}$$

a $*$-binary representation. On printed page 391 of [1], they leave open the possibility that every rational in the range admits a periodic representation and single out $1/3$. Their discussion explicitly allows a period to start after a finite initial segment. The question therefore concerns eventual periodicity, which is the convention used throughout this manuscript.

The quantifier is essential: ruling out one algorithm's periodic output would not rule out other representations. Corollary 2.4 rules out *every* eventually periodic representation of $1/3$, with an arbitrarily long preperiod allowed. Thus it gives a negative answer to that particular universal possibility without changing its meaning. It is an unnumbered question in the cited paper, rather than an established named conjecture or the whole of Erdős #261.

Our auxiliary problem is to find a finite, explicit bound for the ordinary binary tail denominators that can occur when the *weighted* value has a prescribed odd denominator Q. A periodic ordinary binary value has denominator dividing $2^r - 1$, but differentiating the corresponding generating function introduces a second power of this factor. The key question is which prime factors can cancel. The exact identity in Lemma 2.3 controls that cancellation; Theorem 2.6 then propagates the resulting constraints through strictly decreasing primes. This gives an explicit arithmetic obstruction and applies it to the historical question. The tools—geometric series, prime valuations, and multiplicative orders—are classical; no new foundational principle is claimed.

The research began with the weighted-tail viewpoint in manuscript [090] of the supplied corpus [4], inspected at the fixed commit recorded in the bibliography. That corpus is used as a source of research directions, not as an assumption of correctness. Every assertion needed for the present result is proved below.

2 Denominators and eventual periods in weighted binary expansions

For a binary sequence $\varepsilon = (\varepsilon_n)_{n\ge 1}$, write

$$X(\varepsilon) = \sum_{n\ge 1} \frac{n\varepsilon_n}{2^n}.$$

Every $x \in [0,2]$ admits such a representation; the elementary greedy construction is recalled below. We determine an explicit restriction on every eventual period of a representation of a fixed rational number. In particular, $1/3$ has no eventually periodic representation. This answers the specific periodicity possibility discussed by Borwein and Loring on printed page 391 of [1]. It does not settle either of the remaining questions in Erdős Problem #261: universal finite representability of $n2^{-n}$ or continuum multiplicity for some rational value.

The periodic-tail formula used below is classical generating-function algebra. In the closely related recent literature, Wang and Grau Ribas [2, Lemma A.12 and Propositions 7.2/A.15] use this formula and the identification of the primitive binary period with a multiplicative order to obtain a density estimate. The argument here extracts prime valuations from the same formula and propagates them through the prime divisors of multiplicative orders. No claim that the underlying generating-function identity is new is intended.

For an odd prime p, define

$$t_p = \operatorname{ord}_p(2), \qquad w_p = v_p(2^{t_p} - 1).$$

Here v_p is the usual additive valuation, with $v_p(0) = +\infty$. In particular $w_p \ge 1$ and $t_p \mid p-1$.

Lemma 2.1 (Periodic tail formula)**.** *Suppose that $\varepsilon_{N+i} = e_i$ for $i \ge 1$, where (e_i) is periodic of minimal period r, with digits in $\{0,1\}$. Put*

$$M = 2^r - 1, \qquad S = \sum_{i=1}^{r} e_i 2^{r-i}, \qquad T = \sum_{i=1}^{r} i e_i 2^{r-i},$$

$$y = \sum_{i\ge 1} \frac{e_i}{2^i} = \frac{S}{M}, \qquad z = \sum_{i\ge 1} \frac{i e_i}{2^i} = \frac{MT + rS}{M^2}.$$

Then

$$2^N X(\varepsilon) = I + Ny + z, \qquad I = \sum_{n=1}^{N} n\varepsilon_n 2^{N-n} \in \mathbb{Z}. \tag{2}$$

If the periodic word is nonconstant and d is the reduced denominator of y, then $d > 1$ is odd and

$$r = \operatorname{ord}_d(2) < d.$$

Proof. Summing over positions $i + kr$ gives

$$y = \sum_{i=1}^{r} \frac{e_i 2^{-i}}{1 - 2^{-r}}, \qquad z = \sum_{i=1}^{r} e_i 2^{-i} \left(\frac{i}{1 - 2^{-r}} + \frac{r 2^{-r}}{(1 - 2^{-r})^2} \right),$$

which yields the displayed formulas. Separating the finite prefix proves (2). A nonconstant periodic binary word represents a rational y strictly between zero and one. Multiplication by 2^s shifts the ordinary binary digits by s places; its fractional part returns to y precisely when $d \mid 2^s - 1$. Uniqueness of the nonterminating binary expansion therefore identifies its minimal period with the multiplicative order of 2 modulo d. Euler's theorem gives $r \le \varphi(d) < d$. □

Lemma 2.2 (Order lifting)**.** *For an odd prime p and every $a \ge 1$,*

$$\operatorname{ord}_{p^a}(2) = t_p p^{\max(0,a-w_p)}.$$

Moreover, if $t_p \mid r$, then $v_p(2^r - 1) = w_p + v_p(r)$.

Proof. For $u \equiv 1 \pmod p$ and odd p, the binomial theorem gives $v_p(u^m-1) = v_p(u-1)+v_p(m)$. To see this, an exponent coprime to p leaves the valuation unchanged, while raising to the power p increases it by exactly one; iteration proves the identity. Apply it to $u = 2^{t_p}$. Any exponent giving 1 modulo p^a is a multiple $t_p m$, and it works exactly when $w_p + v_p(m) \ge a$. The least such multiple proves the order formula. The final assertion also uses $p \nmid t_p$. □

Lemma 2.3 (Exact denominator amplification)**.** *Under the nonconstant-tail hypotheses of Lemma 2.1, let Q be the odd part of the reduced denominator of $X(\varepsilon)$. For an odd prime $p \mid d$, put*

$$a = v_p(d), \qquad b = v_p(r).$$

If $a > b$, then

$$v_p(Q) = a + w_p. \tag{3}$$

In particular, the odd denominator of $X(\varepsilon)$ is divisible by the square of some prime.

Proof. Write $k = v_p(M)$. Reduction of S/M gives $v_p(S) = k - a$. Since $a > b$,

$$v_p(rS) = k - a + b < k \le v_p(MT).$$

Thus the two numerator summands in $MT + rS$ cannot cancel at their least valuation, and

$$v_p(z) = k - a + b - 2k = -k - a + b.$$

As $p \mid d$, the number t_p divides r. Since $p \nmid t_p$, the lifting-the-exponent identity gives

$$k = v_p(2^r - 1) = w_p + v_p(r/t_p) = w_p + b.$$

Consequently $v_p(z) = -a - w_p$. On the other hand $v_p(Ny) \ge -a$ and $v_p(I) \ge 0$. These terms cannot cancel the strictly smaller valuation of z in (2). Multiplication by 2^N leaves every odd-prime valuation unchanged, proving (3).

Finally, $r < d$ implies $d \nmid r$. Hence there is a prime $p \mid d$ with $v_p(d) > v_p(r)$. For this prime, $v_p(Q) = a + w_p \ge 2$. □

Corollary 2.4 (Squarefree-denominator obstruction)**.** *Let $x \in \mathbb{Q} \cap [0, 2]$, and suppose that the odd part $Q > 1$ of its reduced denominator is squarefree. Then no binary representation*

$$x = \sum_{n \ge 1} n\varepsilon_n 2^{-n}$$

is eventually periodic. In particular, every such representation of $1/3$ is aperiodic after every finite prefix.

Proof. A nonconstant periodic tail contradicts Lemma 2.3. An eventually constant binary sequence gives a dyadic value: a zero tail is finite, while a one tail contributes

$$\sum_{n>N} \frac{n}{2^n} = \frac{N+2}{2^N}.$$

Either is incompatible with $Q > 1$. □

For completeness, existence of the representations appearing in the corollary follows directly from a tail interval. Start with $\rho_0 = x$ and at step $n \geq 1$ choose a digit ε_n for which

$$\rho_n = 2\rho_{n-1} - n\varepsilon_n \in [0, n+2].$$

If $\rho_{n-1} \in [0, n+1]$, exclusion is possible for $2\rho_{n-1} \leq n+2$ and inclusion is possible for $2\rho_{n-1} \geq n$; these conditions cover the interval. Then

$$x - \sum_{j=1}^{n} \frac{j\varepsilon_j}{2^j} = 2^{-n}\rho_n \longrightarrow 0.$$

Thus the conclusion for $1/3$ is not vacuous.

Definition 2.5 (Prime set and period bound attached to a denominator)**.** For an odd positive integer Q, let $\mathcal{P}_Q$ be the smallest set of odd primes with the following properties:

1. it contains every prime p with $v_p(Q) > w_p$;
2. if $q \in \mathcal{P}_Q$, it contains every odd prime divisor of t_q.

This set is finite, since $t_q \mid q-1$ and hence every newly added prime is smaller than its parent. For $p \in \mathcal{P}_Q$ set

$$e_p = \max\left\{0,\ v_p(Q) - w_p,\ \max_{\substack{q \in \mathcal{P}_Q \\ q > p}} v_p(t_q)\right\}, \qquad D_Q = \prod_{p \in \mathcal{P}_Q} p^{e_p}.$$

An empty inner maximum is zero and an empty product is one. Every prime in $\mathcal{P}_Q$ has $e_p \geq 1$. If $D_Q > 1$, put $R_Q = \operatorname{ord}_{D_Q}(2)$.

Theorem 2.6 (A bound for every eventual period)**.** *Let x be rational with reduced denominator $2^s Q$, where Q is odd. Suppose a binary representation of x has a nonconstant eventual period. Let d be the reduced denominator of the ordinary binary value of a periodic tail, and let r be its minimal period. Then*

$$d \mid D_Q.$$

Consequently $D_Q > 1$ and

$$r \mid R_Q.$$

The bound depends only on the odd denominator of x, independently of the numerator, the dyadic denominator, the preperiod, and the chosen representation.

Proof. For each prime $p \mid d$, put $a_p = v_p(d)$ and $b_p = v_p(r)$. We first prove the following alternative:

$$a_p \leq v_p(Q) - w_p \quad \text{or} \quad a_p \leq v_p(t_q) \text{ for some prime } q \mid d \text{ with } q > p. \tag{4}$$

If $a_p > b_p$, Lemma 2.3 gives equality in the first alternative. Otherwise $a_p \le b_p$. The standard order formula at odd prime powers is

$$\operatorname{ord}_{q^a}(2) = t_q q^{\max\{0,a-w_q\}}.$$

Since $r = \operatorname{lcm}_{q|d} \operatorname{ord}_{q^{a_q}}(2)$, taking its p-valuation yields

$$b_p = \max\left\{ \max\{0, a_p - w_p\},\ \max_{\substack{q|d \\ q>p}} v_p(t_q) \right\}.$$

Here $q < p$ contributes zero because $t_q \mid q-1 < p$, and t_p is coprime to p. The first entry is strictly smaller than a_p. Thus $b_p \ge a_p$ forces the second alternative in (4).

Now process the prime divisors of d in decreasing order. The largest one must satisfy the first alternative, so it is a seed of $\mathcal{P}_Q$ and its exponent is at most e_p. For any smaller prime p, the first alternative again makes it a seed with the required exponent bound. In the second alternative, the larger prime q has already been placed in $\mathcal{P}_Q$; since $p \mid t_q$, the closure rule places p there too, and the definition of e_p bounds a_p. This proves $d \mid D_Q$. Multiplicative orders under divisibility then give

$$r = \operatorname{ord}_d(2) \mid \operatorname{ord}_{D_Q}(2) = R_Q.$$

□

Corollary 2.7 (A finite list of possible tail words)**.** *For a fixed odd denominator Q, every ordinary binary value of a nonconstant periodic tail belongs to the finite set*

$$\left\{ \frac{j}{D_Q} : 1 \le j < D_Q \right\}.$$

Thus finitely many periodic binary tail sequences can occur, including their phases. This does not bound the point at which a tail begins.

Proof. The reduced denominator d of the tail value divides D_Q, and the value lies strictly between zero and one. Each such rational has a unique nonterminating, purely periodic binary expansion. □

Example 2.8 (The squarefree obstruction has a real boundary)**.** Although $1/3$ has no eventually periodic representation, $1/9$ does:

$$\frac{1}{9} = \frac{7}{2^7} + \frac{8}{2^8} + \sum_{k\ge 0} \frac{9+2k}{2^{9+2k}}.$$

Indeed the finite prefix contributes $11/128$. The tail begins after $N = 8$ and has periodic word 10, for which $y = 2/3$ and $z = 10/9$. Its weighted contribution is therefore

$$2^{-8}(8y+z) = \frac{29}{1152}, \qquad \frac{11}{128} + \frac{29}{1152} = \frac{1}{9}.$$

Remark 2.9 (Scope and examples)*.* Theorem 2.6 is a necessary condition for periodic representability, not a criterion asserting its existence for each rational numerator. It bounds the period and not the preperiod. For example,

Q	9	25	49	81	121
D_Q	3	5	21	27	55
R_Q	2	4	6	18	20

where the primes occurring here have $w_p = 1$. Thus every nonconstant eventual period at odd denominator 49 divides 6, and at odd denominator 121 it divides 20. Squarefree Q has no seed, giving $D_Q = 1$ and recovering the obstruction above. These conclusions concern a classical periodicity question adjacent to Erdős #261; they provide no proof of infinite branching or continuum multiplicity.

3 Examples and the role of higher prime valuations

The period bound is attained for some denominators. The purely periodic word 0011 has $r = 4$, $M = 15$, $S = 3$, and $T = 10$, so its weighted value is $18/25$. Its minimal period is $4 = R_{25}$. For $Q = 49$, take a zero prefix of length one followed by the repeated word 000011. Here $y = 1/21$ and $z = 38/147$, and hence

$$x = \frac{1}{2}\left(\frac{1}{21} + \frac{38}{147}\right) = \frac{15}{98}.$$

Its minimal period is $6 = R_{49}$. The prime 3 in $D_{49} = 21$ comes from $\mathrm{ord}_7(2) = 3$; it need not divide Q itself. Thus simply retaining the primes dividing Q would lose valid tails.

The use of w_p, rather than the weaker uniform lower bound $w_p \geq 1$, also matters. For the prime $p = 1093$, exact modular arithmetic gives

m	364	182	52	28
$2^m \bmod 1093$	1	1092	27	121

and

$$2^{364} \equiv 1 \pmod{1093^2}, \qquad 2^{364} \equiv 1064432260 \not\equiv 1 \pmod{1093^3}.$$

Trial division proves that 1093 is prime. Since the prime divisors of 364 are $2, 7, 13$, the table proves $t_{1093} = 364$, and the second line proves $w_{1093} = 2$. For $Q = 1093^2$ there is consequently no seed in Definition 2.5; thus $D_Q = 1$.

Corollary 3.1. *No weighted binary representation of* $1/1093^2$ *is eventually periodic.*

Proof. Theorem 2.6 excludes a nonconstant periodic tail, and a constant tail would give a dyadic rational. □

This conclusion is not covered by the squarefree-denominator corollary. The number-theoretic property of 1093 is used only as an explicitly checked example, not as an unproved input.

4 The denominator obstruction in every integer base

Fix an integer $b \geq 2$ and use the full digit alphabet $\{0, 1, \ldots, b-1\}$. A weighted base-b representation is an expansion

$$x = \sum_{n=1}^{\infty} \frac{n\varepsilon_n}{b^n}, \qquad \varepsilon_n \in \{0, 1, \ldots, b-1\}.$$

If x is rational with reduced positive denominator q, write

$$Q = \prod_{\substack{p \mid q \\ p \nmid b}} p^{v_p(q)},$$

where v_p denotes the usual prime-adic valuation. Thus Q is the part of the denominator coprime to b.

Theorem 4.1. *Let x have an eventually periodic weighted base-b representation. If its eventual tail is neither constantly 0 nor constantly $b-1$, then the reduced denominator of x is divisible by p^2 for some prime $p \nmid b$. If the eventual tail is constantly 0 or constantly $b-1$, then $Q \mid b-1$.*

Consequently, if a rational number x has squarefree Q and $Q \nmid b-1$, then none of its weighted base-b representations is eventually periodic.

Proof. Choose $N \geq 0$ such that $\delta_i = \varepsilon_{N+i}$ is purely periodic, and let r be its least positive period. Put

$$M = b^r - 1, \qquad S = \sum_{i=1}^{r} \delta_i b^{r-i}, \qquad T = \sum_{i=1}^{r} i\delta_i b^{r-i}.$$

Summing the geometric series and its derivative gives

$$y := \sum_{i=1}^{\infty} \frac{\delta_i}{b^i} = \frac{S}{M}, \qquad z := \sum_{i=1}^{\infty} \frac{i\delta_i}{b^i} = \frac{MT + rS}{M^2}. \tag{5}$$

Writing the finite prefix as $F = \sum_{n=1}^{N} n\varepsilon_n b^{-n}$, we therefore have

$$x = F + b^{-N}(z + Ny). \tag{6}$$

First suppose that the periodic tail is neither all 0 nor all $b-1$. Then $0 < y < 1$. Its reduced denominator d divides M and satisfies $d > 1$, so $\gcd(d, b) = 1$. The least period of the ordinary base-b expansion of y is

$$r = \operatorname{ord}_d(b) \leq \varphi(d) < d.$$

Indeed, $b^s y - y$ is an integer precisely when $d \mid b^s - 1$; uniqueness of the ordinary base-b expansion of this nonterminating rational identifies these integers s with its periods. In particular, $d \nmid r$. We may therefore choose a prime $p \mid d$ such that

$$a := v_p(d) > h := v_p(r).$$

Let $k = v_p(M)$. Reduction of S/M gives $v_p(S) = k - a$, and hence

$$v_p(rS) = k - a + h < k \leq v_p(MT).$$

The two summands in the numerator of z have distinct p-adic valuations. Thus

$$v_p(MT + rS) = k - a + h,$$

and the exponent of p in the reduced denominator of z is exactly

$$E = 2k - (k - a + h) = k + a - h > a \geq 1.$$

The denominator of Ny has p-exponent at most a, so addition of Ny cannot cancel the larger exponent E in the denominator of z. Moreover, $p \nmid b$, since $p \mid M$. Multiplication by b^{-N} and addition of the finite base-b rational F likewise leave this exponent unchanged. Equation (6) therefore proves that p^2 divides the reduced denominator of x.

If the tail is all 0, then $x = F$ and $Q = 1$. If the tail is all $b-1$, direct summation gives

$$y = 1, \qquad z = \frac{b}{b-1}, \qquad x = F + b^{-N}\left(N + \frac{b}{b-1}\right).$$

All denominator factors outside the primes dividing b consequently divide $b-1$. This proves the two exceptional cases and the stated consequence. □

The extremal-tail exception is necessary. For example, in base 3 the constant digit sequence $\varepsilon_n = 2$ represents $\sum_{n\geq 1} 2n3^{-n} = 3/2$, whose reduced denominator is squarefree. Constant interior digits $1, \ldots, b-2$, when they exist, are covered by the nonextremal case of Theorem 4.1.

Lemma 4.2. *Every real number $x \in [0, b/(b-1)]$ has a weighted base-b representation with digits in $\{0, 1, \ldots, b-1\}$.*

Proof. Set $w_n = n/b^n$ and let

$$H_n = (b-1)\sum_{j>n} \frac{j}{b^j} = b^{-n}\left(n + \frac{b}{b-1}\right) \qquad (n \geq 0).$$

Thus $H_0 = b/(b-1)$, $H_n \to 0$, and $H_n > w_n$ for $n \geq 1$. The intervals

$$[dw_n, dw_n + H_n], \qquad d = 0, 1, \ldots, b-1,$$

overlap and cover $[0, (b-1)w_n + H_n] = [0, H_{n-1}]$. Starting with $R_0 = x$, choose at step n a digit ε_n whose interval contains R_{n-1}, and put $R_n = R_{n-1} - \varepsilon_n w_n$. Then $0 \leq R_n \leq H_n$. For example, the explicit greedy choice

$$\varepsilon_n = \min\left\{b-1, \left\lfloor \frac{R_{n-1}}{w_n} \right\rfloor\right\}$$

has this property. Since $x - \sum_{j=1}^{n} \varepsilon_j w_j = R_n \to 0$, the desired expansion follows. □

In particular, for every prime $p \nmid b(b-1)$ the number $1/p$ lies in the interval of Lemma 4.2 and has at least one weighted base-b representation, but none is eventually periodic. In base 2 one may take $p = 3$.

5 Periodic representations are isolated

For $x \in [0, 2]$, equip the representation fiber

$$\mathcal{F}_x = \left\{(\varepsilon_n) \in \{0,1\}^{\mathbb{N}} : x = \sum_{n=1}^{\infty} \frac{n\varepsilon_n}{2^n}\right\}$$

with the subspace topology inherited from the product topology on $\{0,1\}^{\mathbb{N}}$. A representation is isolated if some finite initial digit block distinguishes it from every other member of $\mathcal{F}_x$.

Theorem 5.1. *Every eventually periodic binary weighted representation is an isolated point of its full representation fiber.*

Proof. Let $\varepsilon \in \mathcal{F}_x$ be eventually periodic. If its tail is constantly 0, then fixing the digits through the start of that tail leaves the smallest possible remaining sum, namely zero. Every subsequent digit is therefore forced. The same argument with the largest possible remaining sum applies to a tail constantly 1. Both cases give isolation.

Otherwise choose $N \geq 0$ and $r \geq 1$ such that the tail after N is periodic with period r and contains both digits. Choose an integer $L \geq \max\{N, 2^{r+1}\}$. Suppose that another representation $\eta \in \mathcal{F}_x$ agrees with ε through index L, and let $n > L$ be their first differing index.

If $\varepsilon_n = 1$ and $\eta_n = 0$, equality of the represented numbers would require the loss $n2^{-n}$ to be compensated by later changes. The largest available compensation is obtained by turning every later zero of ε into a one. Every block of r consecutive digits in this tail contains a one, so in particular

$$\sum_{t\geq 1} \frac{1 - \varepsilon_{n+t}}{2^t} \leq 1 - 2^{-r}.$$

Using also $\sum_{t\geq 1} t2^{-t} = 2$, we obtain

$$2^n \sum_{m>n} \frac{m(1-\varepsilon_m)}{2^m} = \sum_{t\geq 1} \frac{(n+t)(1-\varepsilon_{n+t})}{2^t} \leq n(1-2^{-r}) + 2 < n,$$

where the strict inequality follows from $n > 2^{r+1}$. Compensation is impossible.

If $\varepsilon_n = 0$ and $\eta_n = 1$, the required compensation would instead come from deleting later ones. Since every block of r tail digits contains a zero, the identical estimate with ε_{n+t} in place of $1 - \varepsilon_{n+t}$ again bounds the available compensation strictly below $n2^{-n}$. Both cases are impossible. Thus the cylinder fixing the first L digits meets $\mathcal{F}_x$ only at ε, proving isolation. □

This theorem gives only one direction: eventual periodicity implies isolation. It does not prove that an isolated representation of a rational number is eventually periodic. In particular, excluding eventually periodic representations of 1/3 does not establish that its representation fiber is perfect or uncountable.

6 Literature, scope, and verification

6.1 What has and has not been established

The completed mathematical result is an obstruction to eventual periodicity for all representations of the stated rational numbers, together with a denominator-dependent bound on every possible nonconstant period. The original universal periodicity possibility is false. The statements are unconditional and their proofs do not depend on the finite calculations.

Neither remaining assertion of Erdős #261 is proved here. In the formal-conjectures snapshot [5], these ask whether every positive integer n has a nontrivial finite representation of $n2^{-n}$, and whether some rational number has continuum many representations by increasing positive integers. Aperiodicity is a property of one sequence; it does not imply that a value has many representations. The missing converse discussed in Section 5 must not be silently assumed. We therefore do not present this work as fulfilling the requested conjunction of a still-open named Erdős problem, a complete solution, and certified historical priority.

6.2 Relation to previous methods

The closest identified antecedent is Wang and Grau Ribas [2], specifically Lemma A.12 and Proposition A.15 in version 1. It contains the periodic-tail identity and multiplicative-order reduction used in Section 2. Those ingredients are credited as prior work. The additional conclusions developed here are the exact valuation obstruction and its finite propagation through prime divisors of orders. We use no density theorem from that preprint.

The literature check examined Borwein–Loring, the 2020 paper of Tengely–Ulas–Zygadło [3], and all four available versions of arXiv:2606.24972, together with title, citation, and phrase searches for weighted or $*$-binary expansions, eventual periodicity, 1/3, and squarefree denominators. No earlier identical resolution or period-divisor theorem was found in those sources. This is a bounded search result, not a proof that no prior solution exists. The original source documents that the periodicity possibility was unresolved there in 1990; this manuscript does not certify that it remained unresolved through every intervening publication. In particular, the argument does not establish a historically new foundational principle.

6.3 Exact checks and reproducibility

Appendix A contains the full standard-library Python verifier. It enumerates every nonconstant binary word of lengths 2 through 17, including words with a smaller minimal period. For lengths at most 8, every shift $N = 0, \dots, 2^r - 2$ is tested; for larger lengths it uses $N \in \{0, 1, 2, r, r+1, 2^r - 2, 1000007\}$. It computes the reduced odd denominator exactly, checks $d \mid D_Q$ and $r_{\min} \mid R_Q$, and checks the exact valuation identity whenever its hypothesis holds. The completed run produced

Nonconstant words	262108
Word–shift cases	1917164
Exact valuation checks	3810912
Failed assertions	0

Adding a finite prefix contributes an integer to the scaled expression and cannot change its reduced odd denominator; the dyadic scaling likewise leaves that denominator unchanged. The checker therefore tests the relevant arithmetic directly. The finite counts are consistency checks only. All word lengths, shifts, and prefixes are covered by the proofs, not by an extrapolation of these counts.

A Self-contained exact verifier

Save the following listing as `verify_period_bounds.py` and run with Python 3.9 or later:

```
python verify_period_bounds.py 17
```

It uses only integer arithmetic. The default argument is 17; a smaller argument shortens the diagnostic run. The verifier is embedded in this single LaTeX source and needs no external data.

```
"""Independent exact checks for the proposed weighted-period bound.

No numerical fitting or floating point is used. This tests finite words; the
mathematical proof is the separate descending-prime/valuation argument.
"""
from functools import lru_cache
from math import gcd, lcm
import json
import sys


@lru_cache(None)
def factor(n):
    out = []
    p = 2
    while p * p <= n:
        a = 0
        while n % p == 0:
            n //= p
            a += 1
        if a:
            out.append((p, a))
        p = 3 if p == 2 else p + 2
    if n > 1:
        out.append((n, 1))
    return tuple(out)


def vp(n, p):
    assert n > 0
```

```
    a = 0
    while n % p == 0:
        n //= p
        a += 1
    return a


@lru_cache(None)
def prime_data(p):
    assert p > 2
    t = p - 1
    for q, _ in factor(p - 1):
        while t % q == 0 and pow(2, t // q, p) == 1:
            t //= q
    w = 1
    while pow(2, t, p ** (w + 1)) == 1:
        w += 1
    return t, w


@lru_cache(None)
def bound(Q):
    Qf = dict(factor(Q))
    seeds = {p for p, a in Qf.items() if a > prime_data(p)[1]}
    closure = set(seeds)
    pending = list(seeds)
    while pending:
        q = pending.pop()
        tq, _ = prime_data(q)
        for p, _ in factor(tq):
            if p != 2 and p not in closure:
                closure.add(p)
                pending.append(p)
    D = 1
    period = 1
    exponents = {}
    for p in closure:
        tp, wp = prime_data(p)
        incoming = [vp(prime_data(q)[0], p) for q in closure if q > p]
        ep = max([Qf.get(p, 0) - wp, 0] + incoming)
        assert ep > 0
        exponents[p] = ep
        D *= p ** ep
        period = lcm(period, tp * p ** max(0, ep - wp))
    return D, period, exponents


def run(max_period=17):
    cases = 0
    words = 0
    exact_valuation_checks = 0
    for r in range(2, max_period + 1):
        M = 2 ** r - 1
        for S in range(1, M):
            words += 1
            d = M // gcd(S, M)
            df = dict(factor(d))
            minimal_r = 1
            for p, a in df.items():
                tp, wp = prime_data(p)
                minimal_r = lcm(minimal_r, tp * p ** max(0, a - wp))
            T = sum(i * 2 ** (r - i) for i in range(1, r + 1)
                    if (S >> (r - i)) & 1)
            # Exhaust every shift residue for r<=8. For larger periods use
            # several shifts, including a much larger, irregular one.
            shifts = range(M) if r <= 8 else (0, 1, 2, r, r + 1, M - 1, 10**6 + 7)
            for N in shifts:
                num = M * (T + N * S) + r * S
                Q = M * M // gcd(M * M, num)
                D, period, _ = bound(Q)
                assert D % d == 0, (r, S, N, d, Q, D)
                assert period % minimal_r == 0, (r, S, N, minimal_r, Q, period)
```

```
                for p, a in df.items():
                    b = vp(minimal_r, p)
                    if a > b:
                        _, w = prime_data(p)
                        assert vp(Q, p) == a + w, (r, S, N, p, Q, a, b, w)
                        exact_valuation_checks += 1
                cases += 1
        print(f"r={r}: cumulative cases={cases}", flush=True)
    # An important boundary example: seed 7 must bring in prime 3.
    assert bound(49)[:2] == (21, 6)
    # Explicit Wieferich-prime cases exercise w_p>1 in the construction.
    assert prime_data(1093)[1] >= 2
    assert bound(1093**2)[0] == 1
    assert bound(1093**3)[0] % 1093 == 0
    report = {
        "max_word_length": max_period,
        "nonconstant_words": words,
        "word_shift_cases": cases,
        "exact_valuation_checks": exact_valuation_checks,
        "failures": 0,
        "example_Q49": {"D": bound(49)[0], "period_bound": bound(49)[1]},
        "wieferich_1093": prime_data(1093),
    }
    print(json.dumps(report, indent=2))


if __name__ == '__main__':
    run(int(sys.argv[1]) if len(sys.argv) > 1 else 17)
```

References

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[5] Google DeepMind and contributors, *Formal Conjectures*, `FormalConjectures/ErdosProblems/261.lean`, snapshot inspected 2 November 2026, commit `d995ef5d306637c0a2e489c7e579647168d7df2f`. Used for the precise separation of the numbered problem's parts; not a proof source for the present theorems.